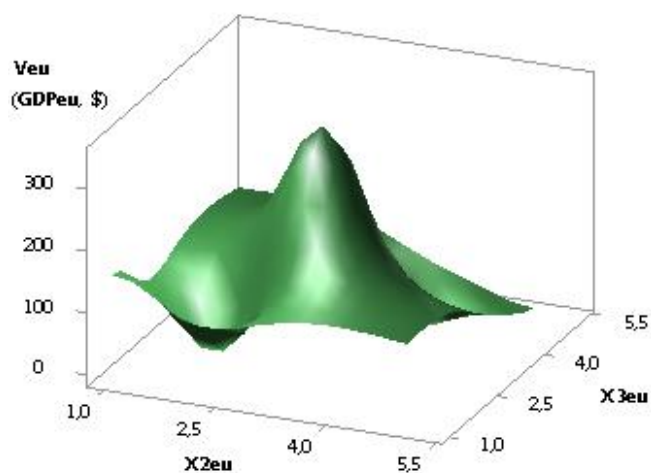


Э. А. ПИЛЬ

НАСТОЛЬНАЯ КНИГА ЭКОНОМИСТА



**САНКТ-ПЕТЕРБУРГ
2025**

СОДЕРЖАНИЕ

INTRODUCTION	10
PART I	
ANALYSIS OF GLOBAL CRISIS	11
1.1. Economic Crises	11
1.11. Classification of Economic Crises	12
1.12. Recession	13
Types of Recession	14
Causes of Recession.....	14
Markers and Signs of Recession	14
Analysis of Recessions	15
The First Economic Crisis in U.S. History	15
The Great Depression and Black Monday	17
The Energy Crisis (1973).....	17
The Iranian Revolution of 1979.....	18
The Global Financial Crisis (2008–2009).....	18
1.2. Financial Crises	18
1.21. Classification of Financial Crises	18
1.3. Currency Crises	19
1.31. Classification of Currency Crises	19
Economic Group of Factors.....	21
Data on the Economic Development of the Country.....	21
Exchange Rate by Purchasing Power Parity.....	22
Gross Domestic Product	22
The Global Currency System.....	23
National and Global Currency Systems	22
The Gold (Gold Standard) Standard	23
The Gold-Exchange Standard (Bretton Woods Currency System).....	24
The Floating Exchange Rate System	24
Currency Exchange Rates and Convertibility	25
Currencies and Their Types	25
Currency Convertibility	26
The Foreign Exchange Market.....	26
Currency Exchange Rates	26
Dynamics of Exchange Rates	28
Currency Corridor	28
Establishment of the Russian Ruble Currency Corridor	30
Currency Regulation	30
Features of Russia's Currency Policy.....	31
The Importance of the IMF in the International Currency Market	32
Main Macroeconomic Indicators	33
Concept of the International Currency Market	33
Features of the Currency Corridor	34
Impact of Central Bank Interventions on the Currency Corridor	34
Seigniorage	34
Important Macroeconomic Indicators of Russia	35
Currency Crises at the End of the Twentieth Century	37
How Oil Affects the U.S. Dollar.....	41
1.4. Banking Crises	44
1.41. Classification of Banking Crises.....	45
1.42. Classification of Banking System Crises by Types and Forms	47
1.43. Mortgage Crisis.....	50
Signs of Mortgage Bubble Formation in the Market	50
1.44. Credit Crisis	50
Causes	50
Simple Lending Conditions	51

Formation of a Bubble	51
Psychological	51
Securities Valuation	52
Consequences.....	52
Historical Perspective	52
1.45. Consequences of Banking Crises.....	52
Worsening financial condition of the state	52
Social consequences.....	53
1.5. Impact of the COVID-19 Pandemic on the US Economy.....	53
Impact of the pandemic on businesses	53
Statistics on company closures	53
One in every 20 franchises in the US closed due to the pandemic	54
Job loss.....	54
The pandemic in the US led to a decrease in bankruptcies.....	55
COVID-19 accelerated retailer bankruptcies in the US	55
Material support for US residents	56
Most in-demand businesses in the US after the COVID-19 pandemic.....	56
1.6. Risks	57
Banking risks: clear definitions and main types	57
1.61. Market Risk.....	57
Stock risk.....	57
Currency risk.....	57
Interest rate risk	57
1.62. Credit Risk	59
1.63. Operational Risk	59
1.64. Liquidity Risk	60
1.65. Deposit Insurance Institutions.....	60
Types of deposit insurance systems	60
Codified deposit protection systems	61
1.66. Country Risk	61
1.67. Legal Risk.....	61
1.68. Reputational Risk.....	61
1.69. Strategic Risk	62
1.7. Gold Reserves of Countries.....	63
1.71. Gold Reserve.....	63
1.72. Sanctions, Blockades, and Seizures	64
1.73. How and Why Central Banks Buy Gold.....	65
How do Central Banks buy gold?	67
1.74. What is the S&P 500 Stock Index?	67
Tradable instruments based on the S&P 500	70
Related indices	70
1.8. Analysis of the US Banking System.....	71
Operating banks in the US	71
Bankruptcies of US banks.....	72
Actions of the US government during the 2008 crisis	74
Recipe for success.....	77
Ben and his helicopter.....	78
1.9. THE WEST MOVES TO NEGATIVE INTEREST RATES.....	78
1.10. Debt Crises.....	79
1.10.1. Classification of Debt Crises.....	82
1.10.2. US Government Debt.....	82
Graham-Rudman-Hollings Act	82
Growth rates in the early 21st century	83
Debt servicing costs	85
1.11. Inflation.....	89
History	89
Causes of inflation	90

Classification of inflation.....	90
By nature of manifestation	90
By growth rate	90
By aggregation level	91
On other grounds	92
Methods of measuring inflation	92
Consumer Price Index.....	92
Other indices	92
Inflation modeling.....	93
Friedman model.....	93
Keynesian model	93
Bruno–Fisher model.....	93
Sargent-Wallace model.....	93
Methods for combating inflation.....	93
Impact of inflation evaluation	94
1.12. Deflation.....	94
Causes of deflation.....	94
Negative consequences of deflation.....	95
Which deflation is not dangerous for the economy	95
Bank credit deflation.....	96
Debt deflation	96
Deflation from money supply.....	96
Credit deflation	96
1.13. Inflation in the US.....	97
History of inflation in the US.....	98
1.14. US Consumer Price Index	99
Methodology	99
Coverage of goods	99
Population and geographical sampling.....	100
Price measurement	100
Housing price measurement (owned homes).....	101
Aggregation	101
Weights.....	102
Different CPI indices	102
Chained CPI for all urban consumers (C-CPI-U)	102
Consumer price index compared to wages	103
History	103
Potential errors in assessment	104
Presumed underestimation of inflation	105
1.15. US Gross Domestic Product	105
1.15.1. Gross Domestic Product	105
Definition	105
Nominal and real GDP.....	105
Gross National Product	106
GDP per capita.....	106
Other definitions	106
Exclusions from GDP calculation.....	106
Calculation methods.....	106
GDP by income.....	106
GDP by expenditure.....	107
GDP by value-added (production method)	107
Criticism.....	107
1.15.2. GDP and GDP per Capita in the US	108
1.16. Analysis of the US Population.....	111
US population by year	111
Mortality and fertility rates in the US	112

Life expectancy in the US.....	112
Literacy rate	112
US population forecast.....	112
1.17. Impact of the COVID-19 Pandemic on the Russian Economy	113
Economic losses incurred by the country	113
Leaders in losses and income for spring 2020 by region (compared to spring 2019)	113
Which businesses were hit hardest.....	113
Tourism	113
Public catering	113
How public catering revenues fell during the lockdown	113
Non-food goods	113
Cultural institutions	114
How entrepreneurs coped with the crisis	114
Optimized costs	114
Launched new directions.....	114
Transitioned to online.....	114
Adjusted to demand	114
Received government support.....	114
1.17.1. Impact of the Pandemic on Russia's External Economy	115
1.17.2. Global Chaos.....	115
1.17.3. Foreign Trade Turnover and Exports.....	116
1.17.4. Imports.....	116
1.17.5. Post-pandemic Macroeconomics	117
1.18. Analysis of the Banking System of the Russian Empire, USSR, and Russia	118
1.18.1. How the Banking System of the Russian Empire Flourished and Perished.....	118
Credit to the grave.....	119
State Bank of the Russian Empire.....	119
The Magnificent Five.....	120
1.18.2. History of the USSR Banking System	120
Before the war.....	120
During the war	121
From war to perestroika	122
Perestroika and the collapse of the USSR.....	122
Savings banks and Sberbank.....	122
What ordinary citizens could use	123
Checks instead of currency	124
Did the USSR have bank cards?	124
1.18.3. Banking System of Russia	125
Banking infrastructure	126
Perestroika in the USSR.....	126
Russia in the 1990s.....	126
Russia in the 2000s.....	127
Bank statistics in Russia from 1993 to 2024.	127
Key interest rate in Russia	128
Operations of the Central Bank of Russia related to the key interest rate	129
Decisions of the Central Bank of Russia on the key rate	129
Effects of the key rate	129
Neutral key interest rate	130
1.19. Inflation in the USSR and Russia	130
1.19.1. Hyperinflation in Early Soviet Russia	130
1.19.2. Inflationary Processes in the USSR During the "Stagnation" and "Perestroika" Periods.....	132
1.19.3. Analysis of Price Movements and Their Factors During the "Era of Stagnation" (1971–1985).....	132
1.19.4. Causes of Inflationary Processes in the Soviet Economy	134
1.19.5. Analysis of Price Movements and Their Factors During the "Perestroika" Period (1985–1991)	135
1.19.6. Exchange Rate of the Ruble to the US Dollar, 1961–2024.....	137
1.19.7. Index of State Retail Prices on Goods in the USSR.....	138
1.19.8. Fighting Inflation	140

Strategic Measures to Prevent Inflation	140
Anti-Inflation Policy of the Russian Federation	141
1.19.9. Economic Crisis in Russia (1998).....	141
Prehistory	142
Financial Situation in Russia from 1995 to 1998.....	142
State of the Economy Before the Crisis	142
Worsening Financial Situation from Late 1997	143
Technical Default and Depreciation of the Russian Ruble	143
Causes of the Crisis.....	144
Characteristics of Stabilization Measures.....	144
Consequences	145
In Russia.....	145
In Other Countries	146
1.20. Gross Domestic Product of Russia	146
1.20.1. First Calculation of Russia's GDP in the 18th and 19th Centuries	147
1.20.2. GDP of the USSR from 1970 to 1990.....	148
1.20.3. GDP of Russia from 1988 to 2023.....	148
1.20.4. GDP of Russia and the USSR per capita as a Percentage of GDP in the USA.....	149
1.21. Population of Russia	151
1.21.1. Population of Russia from 1000 to 1914	151
Historical Demography of the Russian People.....	152
1.21.2. Population of the USSR from 1917 to 1990	153
1.21.3. Population of Russia from 1991 to 2024	154
Mortality	155
Mortality in Russia has decreased to a minimum since 1991	156
Fertility.....	156
In 2023, Russia had the smallest number of births since 1991	157
Steady Decline	158
Marriage.....	160
Types of Marriage According to Legislation.....	160
Divorce.....	161
Divorce in the Russian Empire	161
Divorce in the USSR.....	161
Divorce in the Russian Federation	161
Divorce in Court.....	161
Why Are There So Many Small Banks in the US?.....	162
A Major Problem with Small Banks	162
How Small Banks Shaped the Economy.....	163
But Small Banks Can Still Get into Trouble.....	163
1.22. Options for a Country to Overcome an Economic Crisis.....	163
1.22.1. Options for the US to Overcome an Economic Crisis	164
How Roosevelt Managed to Pull America Out of the "Great Depression" of the 1930s.....	164
How the US Overcame 13 Economic Recessions Since World War II	165
1.22.2. Options for Russia to Overcome Economic Crises in 2008 and 2014.....	168
How Russia Dealt with the Crises of 2008 and 2014	168
How Russia Dealt with the Crises of 2008 and 2014	168
Russia's Exit from the 2008 Crisis	168
Anti-Crisis Policy of the Russian Federation during the 2008 Global Economic Crisis	168
The 2014 Economic Crisis	171
1.22.3. Main Directions for the Country's Development	171
One of the National Goals – Russia's Achievement of 4th Place in the World by GDP Volume ..	172
What Does Technological Leadership in Russia Entail?	172
Among the national goals is ensuring technological independence and the formation of new markets for unmanned aerial systems	172

PART II

2.1. Development of Economic Crises and Their Analysis.....	174
2.1.1. Comparative Analysis of the Theories on the Origins of the Modern Financial-Economic Crisis by M. Khazin and N. Taleb.....	174
CONCLUSIONS.....	179
2.2. Theory of Economic Crises	179
2.2.1. Construction of a Table for Economic and Population Growth Scenarios	179
2.2.2. Scenarios for the Development of Variables e_t and p_t	185
2.2.3. Classification of Development of Variables e_t and p_t	208
2.2.4. Classification of Countries Based on the Area Occupied by Their Economy and Population	209
CONCLUSIONS.....	212
2.2.5. Scenarios for the Development of Macroeconomics After the Impact of External and Internal Pressures....	212
First type of deformation, when all three economic shells remain unchanged.....	215
Second type of deformation, when all three economic shells expand	216
Third type of deformation, when all three shells contract	221
Fourth type of deformation, when the shells of large and medium businesses expand, and the small business shell contracts.....	223
Fifth type of deformation, when the large business shell expands, and the medium and small business shells contract	223
Sixth type of deformation, when the shells of large and small businesses contract, and the medium business shell expands.....	223
Seventh type of deformation, when the shells of large and medium businesses expand, and the small business shell remains unchanged.....	224
Eighth type of deformation, when the shells of large and medium businesses contract, and the small business shell remains unchanged	224
2.2.6. Single Theoretical Volume of the Economic Shell	226
2.2.7. Scenarios for Calculating the Volume of a Spherical Economic Shel	227
Subclasses Without Cracks.....	228
2.2.8. Scenarios for Calculating the Surface Area of a Spherical Economic Shell	231
Subclasses Without Cracks.....	232
CONCLUSIONS.....	237

PART III

APPLICATION OF AUTOMATIC CONTROL THEORY FOR A COUNTRY'S EXIT FROM ECONOMIC CRISIS

3.1. Theory of a Country's Exit from Economic Crisis.....	238
3.2. Theory of Automatic Control.....	238
3.3. Classification of Automatic Control Systems	239
3.4. Classification of Adaptive (Self-Tuning) Systems.....	241
Definition of Adaptive Control Systems (Systems with Variable Structure)	242
Principles of Automatic Control Systems (ACS)	243
Mathematical Models of Linear ACS.....	244
Types of Impacts.....	244
Transfer Function for Connecting Links.....	244
Linearization of Systems and Links	245
Controllability, Observability of ACS.....	246
Stability of Linear Systems.....	246
Stability Criteria	247
Stability Margin of ACS	249
Criteria.....	249
Comparative Characteristics of Stability Criteria.....	249
Dynamic Link	250
Typical Dynamic Links	250
Statics and Dynamics of Control	251
Connections Between Dynamic Characteristics	252
3.5. Automatic Control System for an Economic Crisis.....	253

Decisions Made in Russia During the 2008 Crisis	254
3.6. Types of Stabilization Funds and the Need for Their Formation	254
Eurasian Stabilization and Development Fund	255
Analytical Activities	255
Stabilization Fund of the Russian Federation	256
National Wealth Fund of Russia	256
What is the National Wealth Fund of Russia?	256
Main Goals of the National Wealth Fund's Creation	256
Covering the Deficit of the Federal Budget	256
In 2023, the liquid part of reserves decreased by a third	257
CONCLUSIONS	257
REFERENCES	258
APPENDICES	276
Appendix 1. History of Economic Crises	276
Appendix 2. List of Recessions in the US from 1785 to 2022	279
Appendix 3. Factors of Financial Crises	281
Appendix 4. Currency Crises in Developing Countries and Examples	283
Appendix 5. Price of Unrefined Oil per Barrel (1861–2023)	288
Appendix 6. Banking Crises by Country	290
Appendix 7. List of Banking Crises from 1763 to 2023.	292
Appendix 8. How Much People Receive During Quarantine in Different Countries	294
Appendix 9. Major Gold Reserves by Country from 1970 to 2022	298
Appendix 10. Gold Reserves by Country in 2023.	300
Appendix 11. S&P 500 Index Values from 1871 to 2024.	302
Appendix 12. Number of Banks in the US from 1934 to 2023	304
Appendix 13. Number of Bankrupt Banks in the US from 1934 to 2023	305
Appendix 14. US National Debt from 1929 to 2023	306
Appendix 15. US National Debt from 1791 to 1912.	308
Appendix 16. Debt Crises in Developing Countries Until 2012	311
Appendix 17. Key Interest Rate of the US Federal Reserve from 1950 to 2024	312
Appendix 18. Inflation in the US from 1929 to 2024	315
Appendix 19. US GDP from 1929 to 2023	318
Appendix 20. US Population from 1900 to 2023.	320
Appendix 21. Summary Table of Population, Banks, and Their Ratio in the US	321
Appendix 22. Marriages, Divorces, and Number of Children in Families in the US	323
Appendix 23. Exchange Rate of the US Dollar by Month (1992–2024)	324
Appendix 24. Key Rate of the Central Bank of Russia by Year	327
Appendix 25. Population of Russia from 1000 to 2024.	329
Appendix 26. Banks, Population, and Their Ratio in Russia from 1993 to 2023	330
Appendix 27. Classification of Automatic Control Systems	331

'For me, the word of wisdom is more valuable than gold.'
— *Democritus.*

INTRODUCTION

This book presents extensive material primarily related to economics, which includes an analysis of global economic and financial crises, their classification and description, as well as currency and banking crises. The impact of the COVID-19 pandemic on the economies of the United States and Russia is discussed. Various types of banking risks and the gold reserves of countries are described. An analysis of the banking systems of the United States and Russia is provided, as well as the reasons for inflation and deflation. Additionally, the book includes an analysis of the GDP and population of the United States and Russia, along with a description of how these two countries overcame economic crises. A large number of illustrations, including colored ones, allow readers to clearly see the changes in the parameters under consideration over the years.

In the second part of the book, the author presents his vision of the theory of economic crises, which expands our understanding of this area of science. This section includes tables and diagrams showing various scenarios of economic development, both for the country and its population. The book also describes the types of deformation of economic structures after they are subjected to external and internal pressures.

In the third part, the author proposes using the theory of automatic control for a country's exit from an economic crisis. This section provides a description of its main characteristics and presents a diagram of the automatic control system for managing an economic crisis. Descriptions of the Stabilization Fund of the Russian Federation and the National Wealth Fund of Russia are also included.

At the end of the book, readers can find 27 appendices, which will be especially useful for students in their coursework, articles, and theses.

The author expresses sincere gratitude to the sponsor of the KAF, whose financial support made the publication of this book possible.

The book is dedicated to my son *Mikhail* and grandson *Stepan*, as well as to the bright memory of my parents and older brother.

With best wishes,

Your humble servant and author